

EXERCISE 2

TUTORAT

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Abstract

This report develops Exercise 2 as an empirical extension of Exercise 1 on LLM-based social simulation platforms. Instead of treating an agent conversation as sufficient evidence, it builds an AG2/AutoGen-compatible simulation of stakeholder conflict around the Digital Markets Act and evaluates that simulation through explicit policy-state variables, round-level metrics, contribution decomposition, and sensitivity analysis. The central argument is that a useful social simulation in digital economics must do more than generate plausible dialogue. It must make actor heterogeneity, policy trade-offs, and movement across rounds inspectable. To meet that requirement, the report models five stakeholder roles - regulator, gatekeeper, indie developer, consumer association, and startup founder - plus a moderator-economist. Each agent is associated with a baseline bias, dimension-specific weights, rhetorical resources, and red lines. Each policy round is represented as a vector across contestability, innovation-entry, privacy-security, and implementation feasibility. The deterministic pilot shows a rise in average support from 3.62 to 4.07, a fall in dispersion from 1.05 to 0.82, and a persistent but narrowed gatekeeper gap from 2.15 to 1.76. The most important movement is not a collapse into artificial consensus: the gatekeeper remains only conditionally supportive at the end, while consumer support increases after safeguards and staged implementation are added. Methodologically, the report argues that the code is valuable because it is inspectable, reproducible, and analytically decomposable, yet it still falls short of external validation against real actors. The exercise therefore supports a balanced conclusion consistent with the critical literature: LLM-based social simulation can be useful for heuristic exploration, structured comparison of stakeholder logics, and disciplined scenario design, but it should not be mistaken for causal evidence about real-world regulatory outcomes.

Keywords: digital economics; Digital Markets Act; AG2; AutoGen; generative agents; social simulation; platform regulation; contestability

1. From Exercise 1 to Exercise 2

Exercise 1 established that recent work on LLM-based social simulation should be understood as a field organized around a set of recurring methodological tensions rather than around a single settled architecture. The reviewed corpus moved from foundational memory-reflection-planning systems to large-scale platform claims, empirical applications, and critical reviews. It also stressed that the field's promises - richer cognition, more natural interaction, and greater behavioral flexibility - are inseparable from new risks: opacity, reproducibility problems, weak calibration, and fragile external validity.

Exercise 2 therefore has to do more than stage an interesting dialogue between agents. To remain faithful to the analytical logic of Exercise 1, it must test whether a concrete multi-agent implementation can produce results that are not only readable but also methodologically interpretable. The simulation is not treated as a theatrical transcript. It is treated as a controlled analytical object in which actor positions, policy shocks, support scores, coalition changes, and robustness checks are linked to one another.

The chosen case is the governance of dominant digital platforms under the DMA. This topic belongs squarely to digital economics because it concerns gatekeepers, bottlenecks, switching costs, self-preferencing, interoperability, entry barriers, and the tension between static quality claims and dynamic market contestability. It is also a highly suitable case for social simulation because the relevant actors do not merely disagree in intensity; they disagree in kind. Regulators, gatekeepers,

developers, consumer organizations, and startups do not evaluate the same policy package through the same objective function.

1.1 Conceptual continuity with the literature review

The continuity with Exercise 1 is direct. The earlier review organized the literature into concept buckets: agent architecture, environment and interaction design, scalability, validation, interpretability, governance, and applications. The present exercise operationalizes those buckets in a reduced but clearer setting. Agent architecture becomes the design of roles, red lines, and scoring logic. Environment design becomes the four-round policy sequence with evidence and safeguard shocks. Validation becomes the question of whether score movements remain interpretable and stable under perturbation. Governance becomes the problem of whether the simulation reproduces real stakeholder conflict or only produces polished, synthetic compromise.

For that reason, the present report keeps the intellectual framing of Exercise 1 while moving into an empirical case. The goal is not to claim that the simulation estimates the real causal effect of the DMA. The goal is to test whether a better designed AG2/AutoGen setup can generate an analytically useful representation of stakeholder conflict in digital economics without hiding its own methodological limits.

2. Literature anchor and theoretical justification

The literature reviewed in Exercise 1 provides the framework for evaluating the present simulation. Generative Agents remains the foundational architecture paper because it introduced the now-classic idea that agents can maintain memories, reflect over them, and plan future behavior through language. More recent platform-oriented papers such as GenSim and AgentSociety push that logic toward larger and more reusable infrastructures. At the same time, empirical works on opinion dynamics and social movements show that LLM-based populations can produce richer interaction than classical agent-based models while still displaying important distortions.

That is precisely where the critical paper by Larooij and Tornberg becomes essential. Their review argues that many generative ABM studies still rely too heavily on the subjective believability of outputs, while validation, calibration, and causal explanation remain underdeveloped. Exercise 2 takes that criticism seriously. It does not assume that a plausible transcript is enough. It introduces an explicit measurement layer so that the conversation can be analyzed as a dataset rather than merely quoted as an anecdote.

2.1 Why the DMA is an especially strong digital-economics case

The DMA is not only a current regulatory framework; it is a compact way to formalize core questions in digital economics. The European Commission describes the DMA as the EU law intended to make digital markets fairer and more contestable, especially where gatekeepers control core platform services and act as bottlenecks between business users and consumers. That language maps directly onto the economic concepts at stake in this simulation: market power, intermediation, default effects, dependence, dynamic innovation, and the bargaining asymmetry between dominant platforms and smaller actors.

The topic is also suitable because the conflict cannot be reduced to a simple pro-regulation / anti-regulation axis. The regulator cares about contestability and enforceability. The gatekeeper emphasizes privacy, security, technical feasibility, and ecosystem integration. Developers and startups prioritize discoverability, entry, and lower dependence. Consumer organizations accept stronger interoperability only if privacy and usability are preserved. A high-quality social simulation should reveal these asymmetries rather than flatten them.

2.2 Why AG2/AutoGen is an appropriate orchestration layer

AG2, the continuation of AutoGen, is appropriate for this assignment because it is explicitly designed for multi-agent collaboration and because its documentation supports group-chat style orchestration, including a round-robin speaker selection strategy. That matters methodologically. In a policy simulation, unconstrained speaker choice can itself become a source of hidden variability. A round-robin scheme does not solve the reproducibility problem, but it at least controls turn order and reduces one degree of arbitrariness.

The code keeps AG2 compatibility as an execution path while also providing a deterministic pilot. This is an important design choice. The exercise becomes more defensible when the report is not entirely dependent on one unrepeatable live run. The deterministic pilot provides a stable baseline for tables, figures, and discussion; the live AG2 path remains available when real model interaction is desired.

3. Research design and code architecture

This section translates the conceptual agenda of Exercise 1 into an empirical design suitable for a bounded policy simulation. Rather than introducing a separate set of questions, the DMA case is constructed as a practical test of the issues already identified in the literature review: the realism and limits of ABM, the additional capacities introduced by LLM-driven agents, the methodological risks that accompany those capacities, the diversity of current platform designs, and the conditions under which an agentic case can claim scientific usefulness. The code architecture is therefore not only a technical scaffold; it is the operational form through which the theoretical concerns of Exercise 1 are examined.

The first design choice follows directly from the ABM literature discussed earlier. A digital-market conflict is modeled through a small but heterogeneous set of agents because the key phenomenon is not mass behavior in the abstract, but structured disagreement among institutions with different incentives, vocabularies, and strategic constraints. This makes it possible to observe, within one controlled scenario, how classical ABM bottlenecks reappear when realism depends on language, persuasion, framing, and conditional bargaining rather than on simple payoff-following rules alone. In this sense, the DMA case is used to test whether richer discursive interaction improves explanatory adequacy while still preserving enough structure for analysis.

The second design choice concerns what LLM-driven agents are expected to add. The simulation is deliberately built so that agents must react in natural language to changing policy packages, retain round-level context, and express stakeholder-specific reasoning about contestability, privacy, implementation costs, innovation, and consumer welfare. This operationalizes the claim from Exercise 1 that generative agents may extend ABM through memory, role-consistent interpretation, and flexible planning. At the same time, the architecture is kept transparent enough to show that these gains do not eliminate older scientific problems; they change their form. Calibration remains limited, reproducibility depends on execution mode, and design assumptions continue to shape outcomes through prompts, weights, shocks, and observability choices.

A third design choice is comparative and methodological. The case is intentionally smaller and more inspectable than large-scale platforms such as GenSim, AgentSociety, S3, or GA-S3, because the objective here is not to claim society-level emergence but to build a notebook-sized experiment that can be read, challenged, and extended. The design therefore emphasizes explicit state variables, a support function, metric decomposition, and transcript–metric comparison. In doing so, it treats evaluation not as a final add-on but as a constitutive part of the simulation itself, in line with the concerns about trust, falsifiability, and governance raised in Exercise 1.

This also clarifies the intended status of the DMA scenario. The simulation is framed as a bounded agentic ecosystem: sufficiently rich to expose negotiation, coalition formation, and argumentative adaptation, yet sufficiently constrained to preserve interpretability. That position matters for the broader trajectory of the project. If a tightly delimited case cannot provide inspectable and defensible evidence, then more open and emergent ecosystems will be even harder to justify scientifically. The present design is therefore a threshold test for later agentic environments: it asks what must already be visible, measurable, and contestable at small scale before exploratory complexity can be treated as analytically meaningful.

For that reason, the code architecture separates three layers that are often conflated in weaker generative-simulation demonstrations: simulation, observation, and evaluation. The first layer governs interaction among stakeholder agents and the evolution of policy packages across rounds. The second records state changes, support scores, coalition labels, and contribution vectors. The third subjects those artifacts to comparison through round-level summaries, decomposition, and sensitivity checks. This layered structure is what allows the results section to answer the conceptual agenda of Exercise 1 without reducing the report to a sequence of abstract methodological claims.

The research design can thus be read as a compact bridge between the literature review and the empirical case study. The DMA scenario does not replace the broader questions posed in Exercise 1; it makes them testable within a concrete environment where digital economics, platform governance, and multi-agent simulation intersect.

Round	Policy change	Analytical purpose
Round 0 - Baseline package	Initial DMA-style remedy without detailed safeguard architecture.	Observe prior conflict and identify the first coalition map.
Round 1 - Evidence shock	Evidence indicates anti-steering restrictions and self-preferencing harmed switching and visibility.	Test whether agents react to contestability evidence rather than repeating priors.
Round 2 - Safeguard package	Interoperability is retained but privacy-by-design and staged implementation are added.	Test whether security concerns can be absorbed into a structured compromise.
Round 3 - Final compromise	Auditable milestones and monitoring are added to the safeguard package.	Measure whether conditional support stabilizes without collapsing disagreement.

Table 1. Four-round policy protocol.

Agent	Institutional position	Primary economic logic	Red line / constraint
EU Regulator	Public authority	Contestability, fairness, enforceability	Self-regulation without verifiable remedies
Gatekeeper Strategist	Dominant platform	Security, ecosystem integration, compliance proportionality	Open-ended interoperability detached from safeguards
Indie Developer	Business user	Discoverability, lower dependence, better margins	Opaque ranking and anti-steering restrictions
Consumer Association	Civil society	Choice, switching, privacy, usability	Interoperability without privacy protections
Startup Founder	Entrant firm	Market entry, access to users, anti-exclusion	Indefinite delay and unilateral platform discretion

Moderator-Economist	Analytical moderator	Frame trade-offs and keep decision points explicit	Not scored
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Table 2. Agent roles, institutional positions, and economic logics.

3.1 Methodological requirements for a defensible simulation

A serious academic simulation cannot rely on conversational fluency alone. It must satisfy a set of methodological requirements that make its outputs readable, inspectable, and open to critique. In this report, five requirements structure the code design: clear representation of actor incentives, multi-dimensional policy states, explicit red lines, contribution tracking, and robustness testing. These requirements matter because the object of study is not simply whether agents can speak plausibly, but whether the analyst can explain why support rises, stalls, or fragments.

Concretely, the simulation needs an explicit representation of agent incentives beyond prose descriptions; otherwise positions cannot be compared systematically across rounds. It needs a multi-dimensional measurement layer so that contestability, privacy, feasibility, and entry effects do not collapse into a single uninterpretable value. It needs a distinction between the policy object and the discursive object, because a persuasive intervention is not the same thing as a policy change. It needs a contribution audit so that the analyst can identify which policy feature moved which actor. Finally, it needs a sensitivity test to check whether the conclusion rests on a narrow and fragile parameterization.

Methodological requirement	Why it matters	How the architecture addresses it
Separation between discourse and measurement	Without it, apparent convergence may be rhetorical rather than substantive.	The simulation stores both transcript records and score records for each agent and round.
Multi-dimensional evaluation	A single score can hide trade-offs between privacy, contestability, feasibility, and entry.	The policy package is represented as a four-dimensional feature vector.
Explicit red lines	Without constraints, agents drift too easily toward synthetic consensus.	Each agent has a role-specific red line that shapes acceptable compromises.
Contribution audit	Analysts need to know which policy feature moved each actor.	The code decomposes support changes by policy dimension.
Robustness check	Scenario conclusions should not rely on one exact parameterization.	A Monte Carlo perturbation layer checks directional stability.

Table 3. Methodological requirements of the simulation design.

3.2 The simulation architecture

The code uses a layered architecture. Agents are formalized as dataclass objects carrying a role, an institutional objective, a baseline position bias, dimension-specific weights, rhetorical resources, and explicit red lines. Policy rounds are also formalized as objects. Instead of being described only in prose, each round is represented as a vector over four dimensions: contestability, innovation and entry, privacy and security, and implementation feasibility.

This change matters because it makes agent movement interpretable. Support no longer changes only because the transcript says so. It changes because the policy package changes in measurable ways, and because each agent weights those dimensions differently. The discourse is therefore linked to a state model rather than floating above it.

Layer	What the code does	Why it matters
1. Configuration layer	Dataclasses define agents, policy rounds, objectives, weights, rhetoric, and red lines.	Reduces prompt vagueness and makes assumptions inspectable.
2. Policy-state layer	Each round is represented as a feature vector on contestability, innovation-entry, privacy-security, and implementation feasibility.	Lets the analyst explain why support changes rather than only observing that it changes.
3. Discourse layer	Text interventions are generated from the dominant dimensions for each agent in each round.	Preserves readable debate while remaining linked to measurable incentives.
4. Measurement layer	Every agent-round record stores score, label, and contribution values by dimension.	Creates a proper dataset instead of an anecdotal transcript.
5. Evaluation layer	The code computes mean support, dispersion, coalition mix, gatekeeper gap, and contribution decomposition.	Moves the exercise from storytelling toward structured analysis.
6. Sensitivity layer	A Monte Carlo perturbation test checks whether the main pattern survives small changes in assumptions.	Improves robustness and makes the conclusions less brittle.
7. Live AG2 hook	A separate function can run a real multi-agent interaction with structured JSON outputs.	Keeps compatibility with AG2/AutoGen while preserving analytical discipline.

Table 4. Code architecture and analytical function.

3.3 Support function, labels, and observability

The deterministic pilot computes a support score on a 1-5 scale. At a conceptual level, the score is obtained by combining each agent's position bias with the weighted contribution of the four policy dimensions. The raw value is then transformed through a bounded function so that all scores remain on a common interpretable scale. This is preferable to assigning scores ad hoc because it makes support trajectories traceable to the modeled policy changes.

The code also converts scores into qualitative labels - oppose, conditional, and support - and stores transcript rows with argument tags and round identifiers. The output of the simulation is thus not a single opaque blob of dialogue but a bundle of linked artifacts: transcript records, score tables, summary metrics, contribution decomposition, and sensitivity runs. That observability is what allows the simulation to be discussed as evidence rather than as a purely literary device.

Analytical component	Substantive gain for the exercise
Structured state	The code now treats each agent as an object with weights, bias, rhetoric, and red lines.
Inspectable scoring	Support is computed from a transparent support function driven by policy features.
Transcript dataset	Every intervention is stored with round, score, label, and argument tags.
Coalition metrics	Oppose / conditional / support counts are computed in every round.
Decomposition	The final shift is broken down by policy dimension for each agent.
Sensitivity test	A 1,000-run perturbation test checks whether the direction of change is robust.
Live AG2 compatibility	A separate execution path allows a real multi-agent run with JSON-like outputs.

Table 5. Main analytical components of the code.

3.4 Deterministic pilot versus live AG2 execution

The notebook deliberately preserves two execution modes. In deterministic pilot mode, the state transition logic and rhetorical outputs are generated from the explicit configuration described above. This makes the results reproducible and easy to document in the report. In live AG2 mode, the system can call real agents through a group-chat manager and ask them to emit structured JSON blocks alongside their natural-language interventions. The live path is therefore compatible with the same measurement logic, but it adds model-dependent variability.

This separation is analytically useful. The deterministic pilot is not presented as a substitute for live interaction; it is presented as a defensible baseline. The live run becomes an additional experimental condition rather than the sole source of evidence.

4. Results

The deterministic pilot produces a richer result set than a transcript-only demonstration. Instead of stopping at a qualitative reading of the dialogue, the analysis compares round-level support, coalition composition, the size of the gatekeeper gap, and the contribution of each policy dimension to each actor's movement. The main pattern is clear: support rises only modestly after the evidence shock, but increases more sharply once privacy-by-design safeguards and staged implementation are added. At the same time, disagreement declines without disappearing. This is important because it indicates structured convergence, not synthetic unanimity.

4.1 Round-level support and disagreement

Table 6 summarizes the main quantitative outputs. Average support rises from 3.62 in the baseline round to 4.07 in the final compromise. Dispersion falls from 1.05 to 0.82, and the range contracts from 2.50 to 1.95. The strongest movement occurs between Round 1 and Round 2, precisely when safeguards and staged implementation are introduced. This means the central mechanism of convergence is not the repetition of pro-regulatory arguments but the modification of the policy package itself.

The coalition structure confirms this reading. In the baseline round, the system contains one opponent, one conditional actor, and three supporters. After the evidence shock, the consumer role moves from conditional to support, while the gatekeeper remains opposed. Once safeguards are added, the gatekeeper no longer opposes outright and becomes conditionally supportive. The final round preserves that structure: four supporters and one conditional supporter. No round ends in full consensus.

Round label	Mean support	Std. dev.	Range	Oppose	Conditional	Support	Gatekeeper gap
Baseline package	3.62	1.05	2.5000004	1	1	3	2.15
Evidence shock	3.68	1.13	2.67	1	0	4	2.33
Safeguard package	4.02	0.84	2.0	0	1	4	1.8
Final compromise	4.07	0.82	1.9500002	0	1	4	1.76

Table 6. Summary metrics by round.

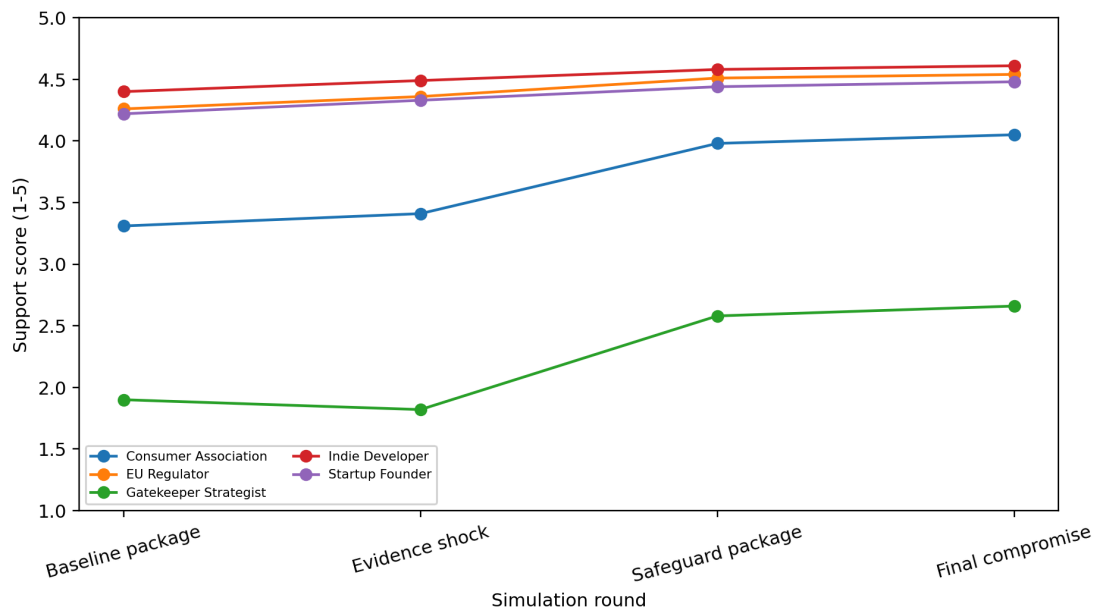


Figure 1. Agent support trajectories across the four policy rounds.

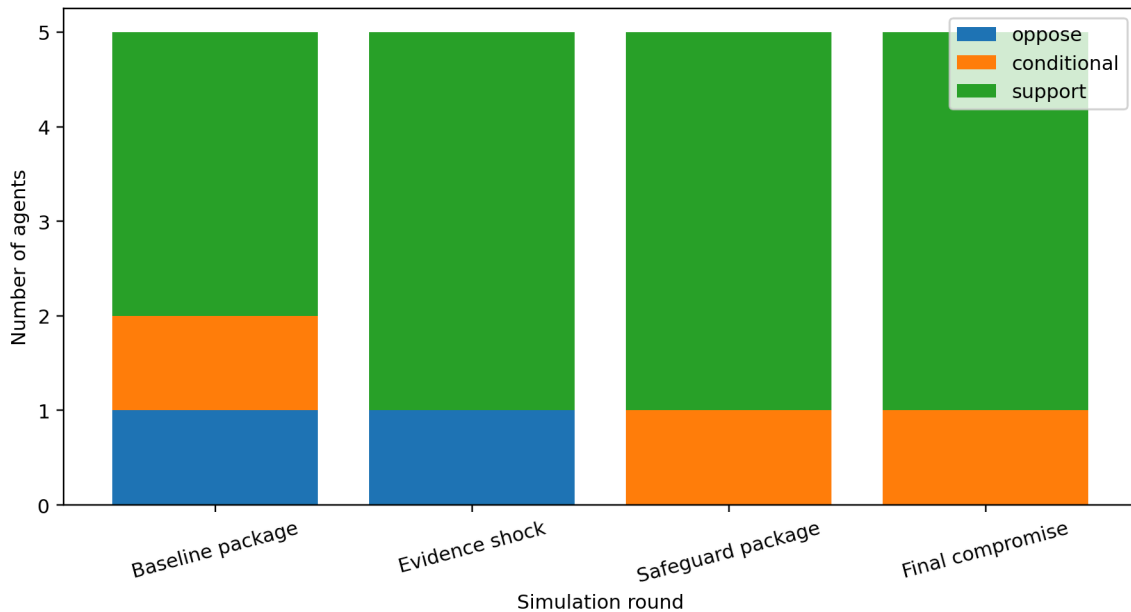


Figure 2. Coalition composition by round: oppose, conditional, and support.

4.2 Agent trajectories and coalition logic

The regulator, indie developer, and startup founder begin with already high support and move upward only moderately. This is analytically plausible. Their institutional logic is already aligned with stronger contestability and lower dependence, so the shocks do not radically change their position. The consumer association behaves differently: it starts at 3.31, close to the threshold between conditional support and support, then rises to 4.05 by the final round. This movement reflects the role's distinctive concern with privacy and usability. It supports openness, but only once safeguards become credible.

The gatekeeper follows the most interesting trajectory. Its support score declines slightly after the evidence shock, from 1.90 to 1.82, because stronger evidence of anti-competitive effects increases pressure for more intrusive remedies. However, once the policy design incorporates privacy-by-design safeguards, staged implementation, and explicit monitoring, the gatekeeper

moves to 2.58 and then 2.66. This still does not amount to endorsement; it is conditional accommodation. That is precisely the kind of realistic conflict the simulation should preserve.

4.3 Contribution decomposition: what changed support?

A major strength of the code is that it can decompose support movement by policy dimension. Moving from the baseline package to the final compromise, privacy and security assurances become the dominant positive driver for the gatekeeper and the consumer association. For the gatekeeper, the delta contribution of privacy and security is +0.292, combined with a +0.200 gain from implementation feasibility. These increases offset, but do not fully erase, the negative effect associated with the contestability dimension. For the consumer association, privacy and security explain most of the upward shift as well (+0.255).

For the regulator, indie developer, and startup founder, the decomposition is more balanced. They benefit modestly from stronger contestability and innovation-entry effects, while also gaining from improved feasibility once the package is staged and monitored. This is a more convincing analytical story than a simple statement that 'support went up'. It shows why support went up, and for whom.

Agent	Contestability Δ	Implementation feasibility Δ	Innovation-entry Δ	Privacy-security Δ
Consumer_Association	0.033	0.125	0.01	0.255
EU_Regulator	0.058	0.1	0.014	0.109
Gatekeeper_Strategist	-0.039	0.2	-0.007	0.292
Indie_Developer	0.052	0.1	0.021	0.073
Startup_Founder	0.046	0.1	0.024	0.073

Table 7. Change in dimensional contribution from the baseline package to the final compromise.

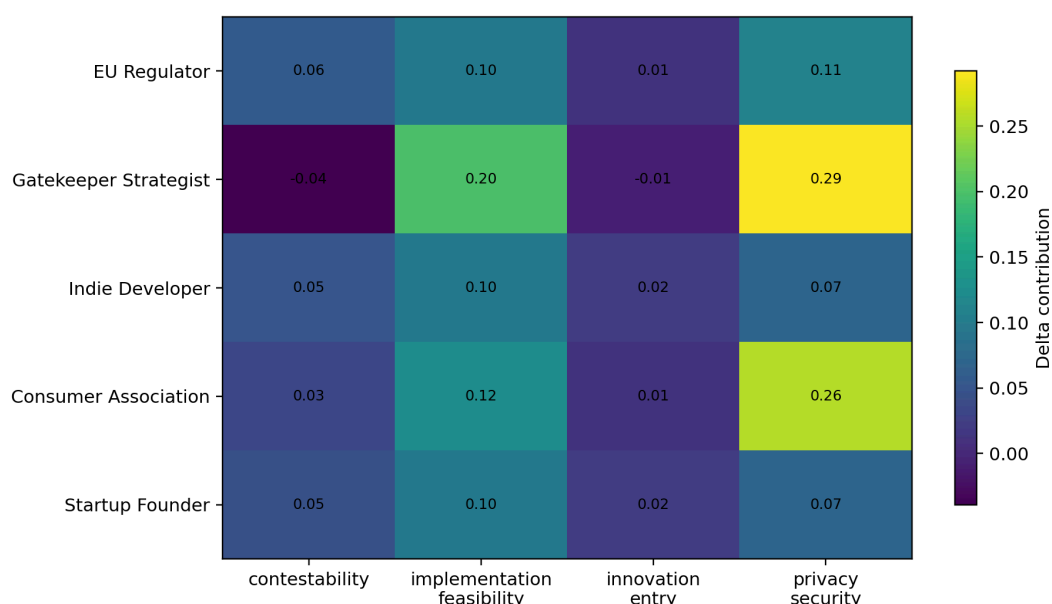


Figure 3. Contribution decomposition by agent and policy dimension.

4.4 Sensitivity analysis

The sensitivity layer is one of the most important parts of the code. Instead of treating the deterministic pilot as exact, the script perturbs biases, weights, and policy-feature values across

1,000 Monte Carlo runs. The purpose is not to estimate probability in a statistical sense, but to test whether the direction of the main conclusion depends on implausibly precise parameter choices.

The results are reassuring at the level of directional robustness. In 100% of perturbation runs, final mean support is higher than baseline mean support. The average change in mean support is +0.451, while the average change in dispersion is -0.236. The probability that the gatekeeper ends at least conditionally supportive is 0.983, whereas the probability of full final consensus is 0.000. This is methodologically useful because it shows that the exercise is not resting on a knife-edge parameterization. The broad pattern - stronger support after safeguards, but persistent disagreement - survives mild perturbations.

Metric	Value
P(final mean support > baseline)	1.0
Average change in mean support	0.451
Average change in dispersion	-0.236
P(gatekeeper ends at least conditional)	0.983
P(full final consensus)	0.0

Table 8. Sensitivity-analysis summary (1,000 perturbation runs).

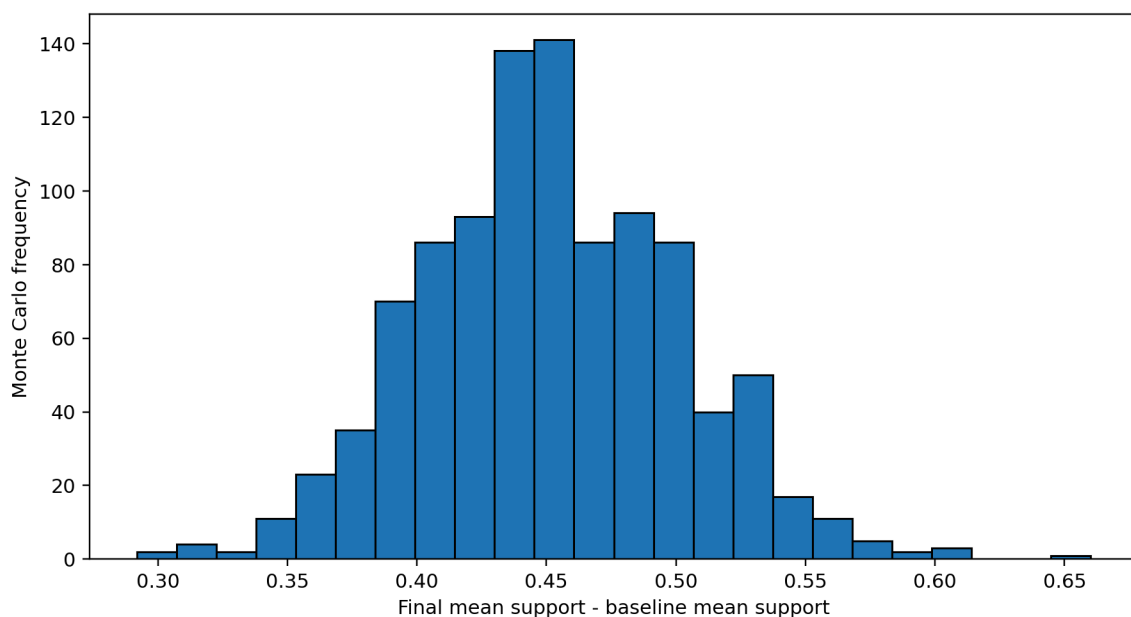


Figure 4. Distribution of the increase in mean support across 1,000 perturbation runs.

4.5 Selected transcript reading

The transcript remains important, but it is now read together with the measurement layer. The regulator's language emphasizes effective contestability, enforceability, and the insufficiency of formal compliance. The gatekeeper emphasizes security and proportionality. The developer and startup founder foreground discoverability, entry, and dependence. The consumer organization conditions support on trust, usability, and privacy. Because these interventions are linked to the dimensional structure of the model, they are more than literary flavor: they help the reader interpret why the scores take the shape they do.

This means the report is not forced to choose between narrative richness and analytical discipline. The transcript gives the simulation social texture; the score and contribution layers prevent that texture from becoming methodologically empty.

5. Deeper analysis of the code

A social simulation cannot be methodologically strong if its code only instantiates a few roles and lets them talk. For a course exercise, that may be visually persuasive, but analytically it remains weak. The code used here turns the simulation into a sequence of linked operations: state specification, policy variation, support calculation, discourse generation, metric extraction, and robustness testing.

The gain is not merely cosmetic. The analyst can point to explicit changes in privacy-security and implementation-feasibility contributions when explaining why the gatekeeper softens its stance, and can show that the consumer association's upward movement is mainly driven by safeguards rather than by a vague drift toward moderation. This is a substantive improvement in interpretability at the level of the modeled system.

5.1 Why discourse-only simulations are weak

A discourse-only setup has three scientific weaknesses. First, it conflates rhetoric with mechanism. Agents may sound convincing even when the underlying logic is unstable or underspecified. Second, it encourages over-reading. A fluent paragraph can invite the analyst to infer strategic change where only stylistic variation occurred. Third, it makes replication difficult because a new live run may produce a noticeably different trajectory with no clear way to compare outputs beyond subjective reading.

The architecture addresses these weaknesses imperfectly but meaningfully. It does so by attaching structured outputs to each step. The support score is not a truth claim; it is an analytical proxy. The argument tags are not a substitute for full semantic parsing; they are a compact way to retain issue salience. The Monte Carlo layer does not validate the model against reality; it checks whether the internal conclusion depends on brittle assumptions.

5.2 What the code captures well

The code captures four things especially well. First, it captures heterogeneity. Different actors do not merely hold different opinions; they apply different weights to the same policy dimensions. Second, it captures conditionality. Support is allowed to move from opposition to conditionality without requiring immediate convergence to endorsement. Third, it captures mechanism. Because the policy package changes dimension by dimension, the analyst can explain which features matter. Fourth, it captures robustness. The sensitivity layer demonstrates that the main pattern is structurally stable under modest perturbation.

These features matter directly for the report's conclusions. The exercise can claim that safeguards and staged implementation are the main bridge between contestability-oriented actors and security-oriented actors in this modeled environment. It can also explain why that bridge remains partial: the gatekeeper still assigns a negative weight to strong contestability measures, so the final coalition stabilizes around conditional support rather than unanimous endorsement.

5.3 What the code still does not solve

Despite these strengths, the code does not solve the main critique raised by Larooij and Tornberg. It still does not establish external validity. The weights, biases, and rhetoric templates are analytically plausible, but they are not calibrated from interviews, platform submissions, regulatory decisions, or systematic corpora of stakeholder statements. The simulation is therefore stronger as a structured thought experiment than as a validated model of real political economy.

Nor does the code eliminate causal ambiguity. Even with a decomposition layer, the support function remains a designed analytical device rather than a discovered law of stakeholder behavior.

The contribution values tell us how the modeled agents move, not how actual firms, regulators, or users must move in the real world. That limitation should be stated plainly, because the field of generative social simulation is often tempted to slide from plausible micro-behavior to inflated macro-inference.

5.4 What a live AG2 run would add - and what it would not add

A live AG2 execution would add discursive richness, unpredictability, and potentially more nuanced reactions to the shock sequence. It could also reveal whether real model interaction produces issue drift, strategic inconsistency, or rhetorical artifacts not visible in the deterministic pilot. Those are valuable observations.

However, a live run would not automatically make the exercise more valid. Without the structured measurement layer, a live transcript remains vulnerable to the same criticism as many generative ABM demonstrations: it may be impressive without being analytically accountable. The correct way to use AG2 here is therefore not to replace the architecture, but to plug real multi-agent interaction into it.

5.5 Formal interpretation of the state-based architecture

The code is best understood as a compact state-based simulation rather than as a sequence of improvised prompts. Each stakeholder is represented by an object that combines institutional identity, a compressed objective, an initial bias toward the policy package, dimension-specific weights, role-specific rhetoric tags, and explicit red lines. This design matters because it separates what belongs to the actor from what belongs to the policy shock. In a purely discursive script, these dimensions tend to collapse into generic conversational behavior. Here, by contrast, the model preserves a distinction between structural incentives and textual expression.

This architecture strengthens the methodological position of the exercise in two ways. First, it makes agent heterogeneity explicit. A regulator and a gatekeeper do not merely 'sound different'; they evaluate the same package through different weight structures and different unacceptable outcomes. Second, it allows the analyst to inspect the simulation at the right level of abstraction. The user can ask whether the weights are defensible, whether the base biases are too strong, or whether the red lines make convergence too easy. Those are contestable modeling choices, but they are contestable precisely because they have been formalized rather than hidden inside a prompt.

State variable	Meaning	Analytical use
role	Institutional identity of the agent	Anchors each actor in a specific political-economic position
objective	Compressed statement of the actor's main interest	Guides argument selection and interpretation
base_bias	Prior inclination toward the policy package	Creates realistic starting asymmetry across actors
weights	Relative valuation of the four policy dimensions	Determines how each policy round affects support
rhetoric_tags	Preferred argumentative vocabulary	Keeps transcript content tied to role-specific issues
red_lines	Conditions the actor refuses to cross	Prevents artificial consensus and keeps bargaining costly

Table 9. Core state variables used to instantiate each stakeholder in the simulation.

The table makes clear why the code can generate politically differentiated movement without requiring a large number of ad hoc exceptions. Support can change over rounds because the policy vector changes, but the meaning of that change depends on the actor's weights and constraints.

That is a stronger design than simply asking agents to 'react' to a scenario, because it makes the source of disagreement inspectable.

5.6 Why the support function matters analytically

The support function is the central bridge between economics and discourse. At each round, the policy package is encoded as a four-dimensional vector: contestability, innovation-entry, privacy-security, and feasibility. Each actor then evaluates the package by applying a simple additive rule that combines a prior bias with weighted responses to those dimensions. This is not a claim that stakeholder preferences are truly linear in the real world. It is a modeling device that forces the evaluation logic into a transparent, auditable form.

This transparency is important for at least three reasons. First, it prevents the transcript from becoming the only evidence of what the simulation 'means'. Second, it allows substantive interpretation: one can test whether support increased because the package improved contestability, because implementation became more credible, or because privacy safeguards reduced perceived risk. Third, it allows controlled comparison with future live AG2 runs. If the same sequence of policy shocks is used in a live run, the analyst can ask whether discursive variation changes the relative importance of the four dimensions or merely changes the wording of positions.

Round field	Meaning	Example in this case study
contestability	Strength of anti-lock-in and anti-self-preferencing remedies	Anti-steering enforcement and ranking neutrality
innovation_entry	Expected effect on complementors and entrants	Better visibility and reduced dependence for developers
privacy_security	Extent to which safeguards are built into interoperability	Privacy-by-design and auditable data-sharing rules
feasibility	Administrative and technical implementability	Staged rollout, milestones, and monitoring architecture

Table 10. Policy-round fields used to encode each regulatory package.

Because these fields are deliberately compact, the model remains intelligible even when it becomes richer. A more elaborate version could introduce interaction terms, threshold effects, or path dependence, but the present formulation already captures a core insight from digital economics: regulatory packages are multidimensional bundles whose acceptability depends on how actors trade off rivalry, innovation, privacy, and administrability.

5.7 Substantive implications for digital economics

The simulation's most meaningful substantive result is not that average support rises. It is that support rises while disagreement remains structured. The gatekeeper never becomes an enthusiastic supporter, even after feasibility and privacy guarantees are strengthened. This matters because digital-platform regulation rarely produces full consensus. A realistic model should preserve enduring conflict where rents, data access, ranking power, and ecosystem dependence remain at stake.

The second important result is that privacy-security operates as a bridge variable rather than as a simple constraint. In many policy debates, interoperability is presented as a pure competition remedy and privacy as a separate legal obstacle. The simulation suggests a different interpretation: carefully designed privacy guarantees can make interoperability politically and institutionally more acceptable to actors who would otherwise resist stronger contestability measures. This is analytically valuable because it mirrors the multidimensional logic of EU digital regulation, where competition, consumer protection, security, and compliance capacity often interact rather than move independently.

A third implication concerns feasibility. The results indicate that implementation credibility has a stabilizing effect across several actors, including those who are otherwise divided on the substance of the package. From a digital-economics standpoint, this is plausible. Regulatory remedies are judged not only by their normative ambition but also by whether they can be monitored, audited, and enforced without producing excessive technical ambiguity. In that sense, feasibility is not a secondary administrative detail; it is one of the channels through which contestability becomes politically durable.

5.8 Recommended live AG2 execution protocol

Even though the deterministic pilot is analytically cleaner, a live AG2 execution remains useful if it is treated as a repeated experiment rather than as a single showcase conversation. A credible live protocol should run the same scenario multiple times under identical role definitions, while recording model identifiers, temperature settings, timestamps, parsed score blocks, and any repair operations triggered by malformed outputs. Without this metadata, observed differences across runs cannot be interpreted properly.

The moderator agent should also be stricter than in a casual demonstration. It should enforce turn order, require a structured output block in every intervention, and trigger a repair instruction when an agent fails to produce the expected fields. This is not a cosmetic issue. Once the simulation is used for analysis rather than entertainment, orchestration reliability becomes part of the object of study. The analyst must know not only what the agents said, but also how often the system had to intervene to keep the exchange analytically usable.

Collected artifact	Why it is needed	Use in analysis
Raw transcript	Preserves discursive texture and strategic framing	Qualitative reading of argument patterns
Structured JSON blocks	Captures scores and issue salience in machine-readable form	Round-level metrics and coalition comparison
Run metadata	Documents model, temperature, and execution conditions	Interpretation of variability across runs
Repair logs	Shows where agent outputs violate the expected schema	Assessment of orchestration reliability
Parsed score tables	Standardizes support values across runs	Comparison with deterministic baseline

Table 11. Recommended artifacts to retain during a live AG2 execution.

The best analytical use of live execution is therefore comparative. The deterministic mode should remain the baseline because it makes assumptions explicit. Live runs can then be used to test whether richer discourse changes support trajectories, coalition structure, and issue salience, or whether it mainly adds stylistic variation. This division of labor protects the exercise from two opposite mistakes: treating a hand-crafted model as if it were already socially realistic, or treating a vivid LLM conversation as if it were automatically valid evidence.

Taken together, these extensions clarify why the code matters analytically and not merely operationally. The simulation is intentionally modest, but it now contains a visible correspondence between stakeholder incentives, policy dimensions, discursive output, and measurable outcomes. That correspondence is what allows the exercise to connect a digital-economics case study to the broader methodological debate on LLM-based social simulation.

5.9 Limits of external validity and calibration

A stronger code architecture does not remove the classical problem of external validity. The agents in this exercise are not estimated from firm data, enforcement case files, consumer surveys, or structured interviews with market participants. They are analytically designed stakeholders. That is

acceptable for a tutorial exercise, but it imposes a strict interpretive discipline: the simulation should be treated as a structured exploratory model, not as evidence that real gatekeepers, startups, or regulators would behave in exactly the same way.

This matters especially in digital economics because platform conflicts are highly context-dependent. The relative importance of interoperability, ranking neutrality, privacy compliance, and implementation cost can vary across search, app stores, ad tech, e-commerce, messaging, and operating systems. A calibration-oriented extension would therefore need sector-specific parameterization. For example, the gatekeeper's weight on privacy-security may be genuinely higher in messaging and identity systems than in marketplace ranking disputes, while the startup founder's weight on innovation-entry may be stronger in app distribution environments than in cloud intermediation.

The same issue applies to the social meaning of the transcript. A plausible sentence produced by an LLM agent does not prove that the underlying institutional actor would use that framing in practice. It only shows that the modeled role can generate coherent language that is consistent with the assigned incentive structure. This is why the transcript should be read as an observable manifestation of the model, not as direct empirical evidence. Calibration would require comparing generated discourse to real policy statements, consultation responses, or public hearing records.

5.10 Robustness and alternative model specifications

The present support function is intentionally linear and transparent, but alternative specifications could produce meaningfully different dynamics. One plausible extension would introduce threshold effects. A gatekeeper might tolerate moderate contestability reforms, then sharply reduce support once interoperability obligations cross a perceived risk threshold. Another extension would add interaction terms, allowing feasibility to amplify or dampen the impact of contestability depending on how implementable the obligations appear. Such variants would not necessarily be more realistic by default, but they would expose whether the conclusions depend on the current additive structure.

A second robustness axis concerns memory and path dependence. In the present notebook, each round updates the policy package and re-evaluates support, but the agents do not accumulate reputational effects, strategic retaliation, or learning from earlier concessions beyond the explicit round design. A richer model could store prior disappointments, detect whether another actor has violated a red line, or increase trust when implementation commitments are honored. This would push the simulation closer to the generative-agent literature reviewed in Exercise 1, where memory and reflection are part of the architecture rather than optional narrative devices.

A third robustness axis concerns population structure. This exercise uses six analytically chosen stakeholders, which is appropriate for a focused regulatory bargaining case. However, one could introduce representative group agents such as multiple developers, several consumer profiles, or competing platform segments. That extension would move the model toward the system-design questions raised by the platform literature, including scaling, orchestration cost, and the tension between richer heterogeneity and reproducibility.

Extension	What changes in the model	Why it matters analytically
Threshold support rule	Support drops or rises non-linearly once a policy dimension crosses a threshold	Tests whether compromise is fragile rather than smooth
Interaction terms	Contestability effects depend on feasibility or privacy safeguards	Captures policy bundling more realistically
Persistent memory	Agents store previous concessions and red-line violations	Introduces trust, retaliation, and path dependence

Sector-specific calibration	Weights differ across app stores, search, messaging, or marketplaces	Improves external validity for digital-economics use cases
Expanded stakeholder populations	Group agents or sampled actor classes replace single representatives	Tests heterogeneity and coalition stability at larger scale

Table 12. Examples of robustness extensions that could be tested in future live executions.

These robustness directions are useful not because the current model is invalid without them, but because they make explicit what kinds of claims the present exercise can support. The current simulation can support claims about the internal logic of a modeled bargaining environment and about the methodological trade-offs between transparency and realism. It cannot, without further calibration, support strong predictive claims about actual enforcement outcomes.

This distinction is also consistent with the critique emphasized in the reference paper: a vivid multi-agent simulation can look socially meaningful long before it becomes scientifically well validated. By adding transparent structure, sensitivity analysis, and a clear robustness agenda, the notebook reduces that risk, but it does not eliminate it.

5.11 Deterministic and live execution as complementary evidence

One of the most important methodological decisions in this exercise is to keep deterministic and live execution analytically separate. The deterministic pilot is not merely a fallback for when an API is unavailable. It is the mode that exposes the structure of the model most clearly: who the actors are, how the policy vector changes, how support is computed, and how disagreement is measured. For a report that must be evaluated on methodological transparency, this clarity is a major strength rather than a limitation.

Live AG2 execution serves a different purpose. It can reveal whether richer dialogue, spontaneous reframing, or unexpected rhetorical coalitions emerge when the structured environment is opened to real model generation. Yet this additional richness should be interpreted as stress-testing the modeled scenario, not as replacing the transparent baseline. In practice, the most defensible design is therefore dual: deterministic execution to stabilize the analytical frame, and repeated live execution to measure how much that frame resists conversational variability.

Execution mode	Main strength	Main risk	Best analytical use
Deterministic pilot	Transparent assumptions and stable measurement	Lower discursive richness	Baseline for interpretation and comparison
Live AG2 run	Richer interaction and emergent framing	Higher variability and formatting noise	Stress-test of the modeled scenario

Table 13. Complementary roles of deterministic and live execution in the evaluation strategy.

This complementary view also helps keep the conclusions proportionate. The report can claim that the simulation is well structured, interpretable, and pedagogically valuable for analyzing digital-platform regulation. It should not claim that the notebook has already solved the validation problem identified in the literature. The value of the exercise lies in making the gap between plausible simulation and empirical validation visible and discussable.

6. Integrated discussion in light of Exercise 1

The value of the DMA case does not lie in offering a separate answer sheet to the literature review, but in making the concerns of Exercise 1 observable inside one bounded empirical design. Once the transcript, support trajectories, coalition movements, dimensional decompositions, and sensitivity checks are read together, the case shows how a generative social simulation can simultaneously extend and complicate the analytical ambitions of classical ABM.

Across the report, a consistent pattern emerges. The simulation becomes more realistic than a purely rule-based model because stakeholders do not merely select options: they justify them, reinterpret them, defend them strategically, and revise them conditionally as the policy package changes. Yet precisely because the model becomes more expressive, the burden of proof shifts toward evaluation, observability, and disciplined limits on what can be claimed. The main contribution of the case is therefore methodological as much as substantive: it demonstrates that in digital-economics settings, the usefulness of LLM-based simulation depends less on vivid conversation than on whether conversational richness is tied to inspectable analytical structure.

6.1 From classical ABM bottlenecks to generative policy interaction

The results clarify why the classical bottlenecks identified in Exercise 1 remain central. In a platform-regulation conflict, realism cannot be reduced to behavior frequencies or simple strategic moves. Coalition formation depends on the language of contestability, implementation feasibility, consumer protection, security, and innovation; agents react not only to incentives but also to how arguments are framed. This confirms that traditional ABM can struggle when the phenomenon of interest is constituted through discourse as much as through action. The DMA case shows that once institutional reasoning becomes part of the object being modeled, discursive richness is not cosmetic but analytically necessary.

At the same time, the simulation makes clear that richer interaction is not equivalent to stronger calibration or external validity. The stakeholder profiles are analytically plausible and grounded in policy logic, but they are not statistically estimated from observed populations. The model therefore supports structured interpretation rather than empirical prediction. The same trade-off appears in scalability. Classical ABMs can scale elegantly with compact rules, while generative simulations become more expensive and harder to audit as conversational depth increases. The code architecture partly addresses this by separating interaction from measurement, but the underlying tension remains: the model gains realism in the domain of argumentation while scientific control becomes more demanding.

This is also where the additional capabilities of LLM-driven agents become visible. The support trajectories and transcript excerpts show differentiated movement by actor type, not because agents are assigned arbitrary labels, but because they can carry institution-specific reasoning across rounds and react flexibly to policy shocks. Memory, language-native interaction, and conditional planning matter here because the debate is not a static preference-revelation exercise. The final compromise emerges through selective concession and reframing, which is precisely the kind of process that Exercise 1 suggested generative agents might capture better than standard ABM.

However, the results also confirm that these gains do not dissolve the methodological criticisms raised in the literature. They reconfigure them. The simulation becomes more persuasive at the surface level, but that persuasive quality must be disciplined by a transparent support function, decomposition logic, repeated execution, and sensitivity analysis. In other words, the case supports the view that LLM-based ABMs do not simply solve classical problems; they relocate them toward design dependence, bias exposure, execution cost, non-determinism, and weak external validation if those risks are not actively managed.

Seen this way, the empirical findings respond to the first part of the Exercise 1 agenda not through isolated propositions but through a cumulative demonstration: ABM bottlenecks remain real, LLM-driven agents add genuinely useful representational capacity, and the net analytical gain depends on whether those added capacities are embedded in a measurement framework strong enough to keep the model interpretable and contestable.

6.2 Platform design, observability, and scientific usefulness

The case also helps position the present implementation within the platform landscape mapped in Exercise 1. Compared with systems such as GenSim or AgentSociety, this AG2/AutoGen-style design is intentionally narrow, small-scale, and notebook-oriented. It does not attempt to simulate thousands of agents or to claim macro-social emergence. Instead, it clarifies what a modest but inspectable platform can contribute: explicit roles, traceable policy shocks, state-based support updates, and a direct link between transcript-level dynamics and metric-level outputs. The comparison is important because it shows that platform differences are not only differences of scale; they are also differences in what kind of evidence they are able to produce and defend.

This is why the evaluation layer matters so much. A transcript by itself can produce a compelling narrative without generating robust evidence. By contrast, the present design forces the simulation to expose round-level support, disagreement, coalition structure, and the contribution of specific policy dimensions. These measurements remain simplified, but they create a falsifiable surface: one can challenge the weights, rerun the model under perturbed assumptions, compare deterministic and live executions, or question whether the observed convergence is an artifact of design. That is a meaningful step toward the sort of benchmarking logic called for in Exercise 1.

The broader lesson is that trust in generative social simulation depends on auditable scaffolding. Scientific usefulness emerges when behavioral richness is paired with artifacts that can be retained, compared, and criticized: prompts, role definitions, parameter settings, transcripts, score tables, and sensitivity results. Governance enters at this point as well, because the researcher must specify what kind of claim the platform is authorized to make. In the present case, the model can illuminate stakeholder logics and policy trade-offs within a bounded scenario, but it cannot justify strong predictive claims about real DMA enforcement outcomes.

The results therefore support a disciplined view of platform evaluation. What matters is not whether the architecture looks sophisticated in the abstract, but whether its outputs remain inspectable enough to distinguish explanation from plausible storytelling. In this sense, the DMA exercise functions as a small but useful benchmark of the benchmark problem itself: it shows how much analytical structure is needed before a generative simulation becomes more than an interesting demonstration.

That conclusion also links directly back to the code discussion. The state-based architecture, support function, and contribution decomposition are not merely implementation choices; they are the mechanism through which the platform becomes scientifically legible. Without them, the same dialogue could still appear realistic, but the report would lose most of its capacity to answer the methodological concerns inherited from Exercise 1.

6.3 Bounded validity and the threshold for more emergent agentic ecosystems

A final implication concerns the status of bounded versus more open agentic environments. The DMA case is defensible precisely because it remains narrow enough to specify its agents, environment, policy shocks, and observables. This boundedness does not weaken the exercise; it is what makes interpretation possible. Because the model states what is being simulated, what counts as movement, and how movement is measured, its conclusions can remain proportionate to its design. That is the condition under which a case study begins to qualify as social simulation rather than as a vivid but weakly testable thought experiment.

This threshold is crucial for any later move toward more emergent ecosystems such as OpenClaw or Moltbook. If a tightly delimited policy scenario still requires explicit observability, repeatable measurement, and narrow causal claims, then environments with looser boundaries and more open-ended interaction will require those safeguards even more urgently. Complexity or surprise

cannot substitute for evaluability. What matters is whether the system exposes enough structure to permit comparison across runs, sensitivity analysis, and bounded interpretation.

The DMA case therefore provides an implicit criterion of validity for future work. An agentic ecosystem becomes analytically credible when it combines exploratory richness with three safeguards: reconstructible interaction traces, a clear account of what its outputs are evidence of, and an evaluation layer capable of resisting purely aesthetic plausibility. These are not optional refinements added after the fact. They are preconditions for claiming that the observed social dynamics matter scientifically.

Taken together, the integrated findings of this report suggest a coherent answer to the broader Exercise 1 agenda. Generative social simulation is most valuable when it is treated neither as a replacement for classical ABM nor as a theatrical display of artificial conversation, but as a hybrid research instrument: more expressive than rule-only models, yet scientifically useful only when its expressive power is disciplined by transparency, measurement, and restrained inference.

That is ultimately what the DMA case contributes. It shows that a carefully bounded digital-economics scenario can function as a serious empirical continuation of the literature review, not because it resolves every open question, but because it turns those questions into observable tensions inside a concrete simulation: realism versus control, expressiveness versus calibration, platform sophistication versus inspectability, and exploratory richness versus valid social-scientific inference.

7. Conclusions

Exercise 2 combines a digital-economics scenario, a structured representation of stakeholder incentives, a transparent support function, transcript-level observability, coalition metrics, contribution decomposition, and a sensitivity check. Together, these elements make the exercise more than a plausible demonstration: they turn it into a disciplined analytical prototype that can be discussed in methodological terms.

Substantively, the results suggest that the modeled conflict around DMA-style remedies is not resolved by abstract pro-market rhetoric or by generic compromise. It is resolved only partially, and specifically through safeguards that change the perceived privacy-security and feasibility profile of the package. That is why the gatekeeper moves from opposition to conditional support rather than to full support, and why the consumer role strengthens its support after Round 2. The final state is therefore more realistic than a unanimous resolution: four supporters and one conditional supporter, with disagreement narrowed but not erased.

Methodologically, the exercise supports a careful conclusion aligned with the critical literature. LLM-compatible social simulation can be useful for organizing heterogeneous stakeholder logics, testing scenario design, and making policy disagreement analytically legible. But even a richer code base does not by itself establish empirical validity. The exercise should therefore be read as a transparent, inspectable, and pedagogically strong simulation of regulatory conflict in digital economics - not as causal proof of what the DMA will do in the real world.

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Appendix A. Selected transcript excerpts

Round	Speaker	Label	Excerpt
0	EU Regulator	support	Market contestability requires effective anti-steering and ranking neutrality. Entry and innovation improve when dependence on bottleneck platforms falls.

0	Gatekeeper Strategist	oppose	Not every ranking outcome is anti-competitive; integration can generate efficiencies. Forced openness can increase security vulnerabilities and reduce user trust.
0	Indie Developer	support	When ranking and steering are biased, small developers cannot compete on merit. Lower dependence creates room for experimentation and differentiated business models.
0	Consumer Association	conditional	Consumers benefit when switching costs fall and alternatives become visible. Choice without trust is not a good bargain; safeguards are essential.
0	Startup Founder	support	A contestable platform environment is a precondition for credible entry. Entrants invest less when they expect to be copied, buried, or blocked.
1	EU Regulator	support	Formal compliance is insufficient if downstream visibility remains distorted. Entry and innovation improve when dependence on bottleneck platforms falls. The evidence shock increases the case for stronger enforceable remedies.
1	Gatekeeper Strategist	oppose	Not every ranking outcome is anti-competitive; integration can generate efficiencies. Forced openness can increase security vulnerabilities and reduce user trust.
1	Indie Developer	support	When ranking and steering are biased, small developers cannot compete on merit. Lower dependence creates room for experimentation and differentiated business models.
1	Consumer Association	support	Consumers benefit when switching costs fall and alternatives become visible. Dynamic competition can improve quality and price discipline for users.
1	Startup Founder	support	A contestable platform environment is a precondition for credible entry. Entrants invest less when they expect to be copied, buried, or blocked.

Appendix Table A1. Sample transcript rows from the deterministic pilot.

Appendix B. Replication bundle

The replication bundle contains the Python script, the notebook, score tables, transcript records, summary metrics, contribution decomposition, and the four figures used in this report.

The notebook can be run in deterministic mode to reproduce the report tables and figures. It can also be switched to live AG2 mode once the appropriate package and API credentials are configured. In the live path, the recommended practice is to archive the raw transcript and the structured JSON outputs for each turn, then feed those artifacts back into the same evaluation layer used in the deterministic pilot.